Bal Bharati Hackathon 3.0

Idea Submission

AgriSense

Problem Statement Title	*Fragmented and Inefficient Farm Management for Small and Medium-Scale Farmers*
Theme	*Sustainability & Green Innovation*
Team Name	*AgriMitra*
Team Leader	*Inaya Sharma*
Unit Name	*Bal Bharati Public School, Dwarka, New Delhi, India*
Team Size	*5 Members*

Your Idea

IDEA / SOLUTION :

AgriSense — An AI-powered Smart Farming Ecosystem
A unified web application integrated with IoT hardware.
Monitors real-time farm conditions using IoT sensors.
Uses AI, weather data and crop requirements to generate personalized recommendations.
Automatically controls irrigation based on soil moisture.
Provides weather intelligence, satellite farm monitoring and smart alerts.
Supports multilingual and voice-based assistance.
Enables farmers to make data-driven decisions from a single dashboard.

Problem Resolution :

Farmers currently depend on multiple sources for weather, crop guidance, irrigation, market information and government schemes.
Manual observation can make early detection of field problems difficult.
Inefficient irrigation can lead to unnecessary water usage.
Existing smart-farming solutions may be expensive or complex for small and medium-scale farmers.
AgriSense brings monitoring, AI recommendations, weather intelligence and automation onto **one accessible platform**.

TECHNICAL APPROACH

Title: System Architecture & Workflow
Content:
A high-level architecture diagram showcasing key components:

- Frontend (User Interface): Where candidates interact with the interview system.
- Backend: Where the processing happens — data processing, response analysis, scoring.
- Cloud Services: Hosting and scaling (mention platforms like AWS, Google Cloud, Azure if applicable).

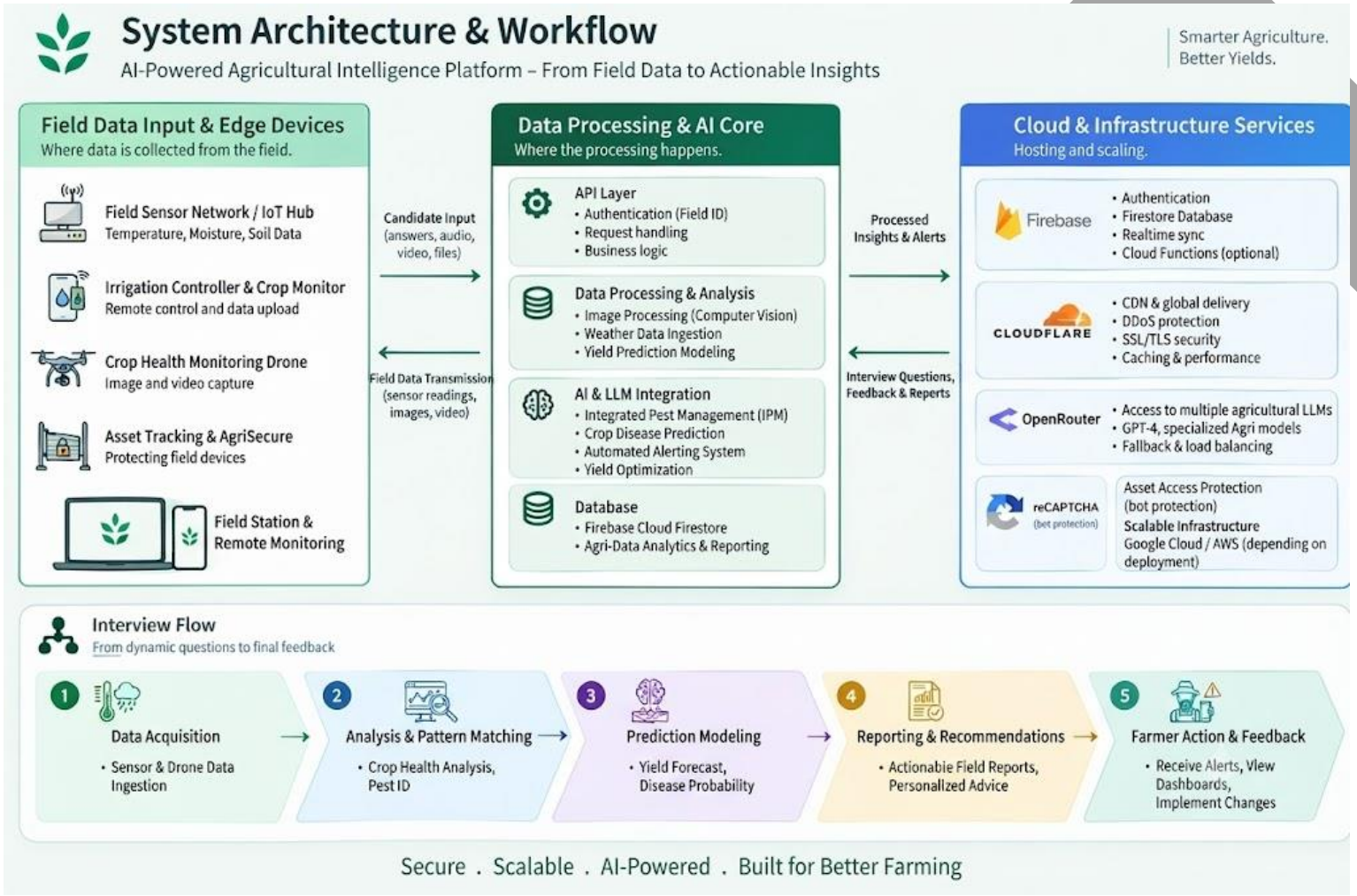
Clearly show the flow of data:

Candidate inputs → if any AI processing → Response evaluation → Interview feedback generation.

Highlight the interview flow:

Example: Dynamic Question Generation → Candidate Response Collection (Text/Voice/Video) → AI-driven Evaluation → Report/Score Generation.

PROCESS FLOW ARCHITECTURE



Remove - IDEA TITLE

Feasibility and Viability

FEASIBILITY OF THE IDEA

- Technical feasibility — Uses existing IoT, AI, cloud and web technologies to collect and analyse real-time farm data. Automated irrigation connects AI-based insights directly to farm operations
- Operational feasibility —A single, simple dashboard combines monitoring, recommendations, weather and alerts. Multilingual and voice assistance makes the platform easier for farmers to use.
- Resource feasibility — The solution uses accessible technologies including IoT sensors, AI, cloud services and weather data. It can be developed and expanded in stages.

POTENTIAL CHALLENGES & RISKS

Sensor accuracy: Incorrect readings may affect recommendations.
Connectivity: Rural areas may have unreliable internet access.
User adoption: Farmers may need time to adapt to digital farming tools.

STRATEGIES TO OVERCOME THEM

Sensor accuracy: Regular calibration and data validation.
Connectivity: Store data and maintain functionality during temporary connection loss.
User adoption: Simple interface with multilingual and voice-based guidance.

Impact and Benefits

Social Impact

- Makes smart farming technology more accessible to small and medium-scale farmers.
- Provides farming guidance through a single platform.
- Multilingual and voice assistance can improve accessibility for farmers from different language backgrounds.
- Helps farmers make decisions using real-time field information instead of relying only on manual observation.

Economic Impact

- Reduces manual effort through automation.
- Helps optimize irrigation based on actual soil moisture.
- Provides data-driven recommendations for farm management.
- Cloud-based architecture can support farms of different sizes

Environmental Impact

- Smart irrigation can reduce unnecessary water wastage.
- Real-time monitoring enables more efficient resource management.
- Data-driven farming supports more sustainable agricultural practices.

BENEFITS OF THE SOLUTION

- For the end user / citizen — [direct benefit in plain terms]
- For the government / industry partner — [operational or policy benefit]
- Scalability — [how the solution can grow beyond the initial use case]

Benefits

For the user

- One platform for farm monitoring and decision-making.
- Real-time field condition monitoring.
- Personalized AI-powered crop recommendations.
- Automated irrigation.
- Live weather intelligence.
- Voice alerts and multilingual support.
- Satellite-based farm monitoring.
- Smart notifications for critical field conditions.

For the Government

- Centralized farm data and historical records.
- Scalable cloud-based infrastructure.
- Better visibility of farm conditions.
- Potential for integration with future agricultural services.

Scalability

AgriSense can scale from individual farms to larger agricultural ecosystems through:
IoT sensors → Cloud infrastructure → AI analysis → Multi-farm dashboards → Multilingual support

Research and References

CITE THE SOURCES THAT SUPPORT YOUR IDEA

- **OneSoil:** https://onesoil.ai/
- **Granular (by Corteva Agriscience):** https://www.corteva.com/us/products-and-solutions/digital-solutions/granular-insights.html
- The Digital Agriculture Mission & AgriStack (Approved Sept 2024)
- Krishi Decision Support System (K-DSS)
- National Pest Surveillance System
- The Food and Agriculture Organization (FAO) report "Scientific review of the impact of climate change on plant pests,"

Explain The Technology Stack

Explain The Technical Architecture and the Technology Stack

The application's technical architecture is built on a robust, modern, and modular stack designed to deliver high performance, rich interactivity, and seamless cloud integration. On the frontend, the architecture relies on clean HTML5, advanced CSS3 styling utilizing custom properties, Flexbox, and Grid layouts, coupled with lightweight Vanilla JavaScript to manage core application logic, state, and user interactions without the unnecessary overhead of heavy frontend frameworks. For high-performance 3D graphics and immersive visualization, Three.js (r128) is integrated directly via CDN. Typography and visual hierarchy are carefully enhanced through Google Fonts, specifically utilizing Sora for clean user interfaces and IBM Plex Mono for precise technical data and numerical readouts. The backend and cloud infrastructure leverage the comprehensive Firebase ecosystem—incorporating Firestore, Realtime Database, Authentication, and App Check security to ensure secure data handling and user management workflows. Furthermore, the application extends its functional capabilities through a wide array of specialized third-party services and APIs, including Google Generative Language and OpenRouter AI for intelligent machine learning features, Esri ArcGIS for dynamic geospatial map rendering, Open-Meteo APIs for real-time weather forecasts and geocoding data, as well as Cloudflare Insights and RSS2JSON for telemetry analytics and feed management. This harmonious blend of lightweight client-side components and powerful cloud-backed APIs creates a resilient, highly scalable, and responsive technical ecosystem tailored to deliver a seamless user experience.

Team Member Details

Student's Details

SL	Student ID	Name	Class	DOB	Email	Phone Number	Gender
1	4420-15	Inaya Sharma	X-B	7/4/2011	Officially.inayasharma@gmail.com	7011688850	Female
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Mentor Details

SL	Name	UID	Address	DOB	Email	Phone Number	Gender
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